

SAWI AGROMET, Thailand

WMO: 485200

Lat: 10.3306N

Lon: 99.0925E

Elev: 13

StdP: 101.16

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 19.7	(c) 20.8	(d) 17.3	(e) 12.4	(f) 21.9	(g) 18.5	(h) 13.4	(i) 22.5	(j) 8.2	(k) 29.3	(l) 6.2	(m) 28.9	(n) 0.1	(o) N/A	(p) 0.293

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 9.3	(c) 35.3	(d) 26.2	(e) 34.5	(f) 26.0	(g) 33.8	(h) 25.9	(i) 27.0	(j) 32.4	(k) 26.7	(l) 31.9	(m) 26.5	(n) 31.5	(o) 1.4	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.8	21.2	28.7	25.6	20.8	28.4	25.2	20.4	28.0	85.1	32.1	83.7	32.0	82.7	31.9	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
5.3	4.3	3.1	DB	17.9	36.5	1.4	0.9	16.9	37.2	16.1	37.7	15.3	38.2	14.3	38.8	
			WB	16.9	27.6	1.5	0.4	15.8	27.9	14.9	28.2	14.1	28.4	13.0	28.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	26.0	26.7	27.9	28.9	29.0	28.4	27.8	28.1	27.8	27.5	26.9	26.1
	DBStd	1.50	1.30	0.92	1.21	1.23	1.29	1.18	1.04	0.99	1.09	1.06	1.15	1.31
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6425	496	468	554	568	588	552	553	562	534	543	506	500
	CDD18.3	3383	237	234	296	318	329	302	295	304	284	285	256	242
	CDH23.3	33969	1997	2252	3191	3648	3654	3263	3018	3163	2820	2718	2262	1983
CDH26.7	12809	629	889	1452	1700	1549	1242	1041	1138	963	920	699	586	
Wind	WSAvg	0.6	0.7	0.6	0.6	0.5	0.4	0.5	0.6	0.6	0.5	0.4	0.6	0.8
Precipitation	PrecAvg	2331	65	40	108	99	234	253	285	297	272	294	248	136
	PrecMax	2898	311	204	510	230	451	447	481	752	442	614	534	494
	PrecMin	1741	4	0	1	1	104	84	104	120	140	86	55	0
	PrecStd	299	61	49	137	69	90	83	97	131	77	116	126	127
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.3	33.2	35.6	36.7	36.2	35.1	34.0	34.1	34.3	33.8	32.9	32.7
		MCWB	25.3	25.1	25.7	26.6	26.3	26.2	25.9	25.9	26.0	26.0	25.6	25.6
	2%	DB	31.6	32.2	34.4	35.5	35.0	34.1	33.0	33.2	33.2	32.8	32.1	31.7
		MCWB	25.3	25.2	25.8	26.3	26.2	26.1	25.8	25.8	25.8	25.8	25.5	25.5
	5%	DB	30.9	31.7	33.4	34.6	34.0	33.1	32.2	32.6	32.4	32.1	31.4	30.9
		MCWB	25.0	25.2	25.7	26.1	26.0	26.0	25.7	25.7	25.6	25.6	25.5	25.3
	10%	DB	30.0	31.0	32.5	33.5	33.0	32.1	31.3	31.7	31.3	31.2	30.6	29.9
		MCWB	24.7	25.0	25.6	26.0	26.0	25.8	25.5	25.5	25.5	25.4	25.3	24.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.1	26.2	26.7	27.7	27.2	27.1	26.9	26.7	26.7	26.8	26.6	26.7
		MCDB	30.4	31.0	32.8	33.5	32.5	32.3	31.2	31.8	31.8	31.3	30.1	30.8
	2%	WB	25.6	25.7	26.3	27.1	26.7	26.6	26.2	26.2	26.2	26.3	26.1	26.0
		MCDB	29.8	30.6	32.1	32.8	32.0	31.7	30.8	31.2	31.2	30.7	30.0	29.9
	5%	WB	25.3	25.5	26.0	26.7	26.5	26.3	26.0	25.9	25.9	25.9	25.7	25.6
		MCDB	29.4	30.3	31.5	32.2	31.6	31.1	30.6	30.8	30.6	30.1	29.5	29.3
	10%	WB	25.0	25.2	25.7	26.3	26.2	26.0	25.7	25.6	25.6	25.6	25.5	25.2
		MCDB	28.9	29.8	30.9	31.7	31.1	30.7	29.9	30.0	29.9	29.5	29.1	28.8
Mean Daily Temperature Range	5% DB	MDBR	8.1	9.3	9.6	9.3	8.2	7.2	6.7	7.1	7.0	7.2	6.7	7.2
		MCDBR	9.3	10.3	11.2	11.1	9.9	9.1	8.5	8.8	8.9	8.9	8.7	8.8
		MCWBR	3.6	4.0	3.8	3.0	2.2	2.3	2.4	2.4	2.4	2.6	2.9	3.3
	5% WB	MCDBR	8.3	9.3	10.2	10.1	8.9	8.0	7.6	7.9	8.0	8.1	7.7	7.8
MCWBR		3.4	3.7	3.6	3.2	2.3	2.3	2.4	2.3	2.5	2.7	2.8	3.1	
Clear-Sky Solar Irradiance	taub		0.490	0.507	0.513	0.479	0.451	0.450	0.449	0.443	0.428	0.458	0.480	0.481
	taud		2.170	2.118	2.145	2.255	2.375	2.412	2.398	2.407	2.440	2.335	2.243	2.214
	Ebn at Noon		819	820	820	839	846	837	841	858	880	853	822	818
	Edh at Noon		149	162	160	142	123	117	119	120	117	129	138	140
All-Sky Solar Radiation	RadAvg		4.82	5.54	5.79	5.83	4.94	4.61	4.41	4.54	4.62	4.35	4.26	4.34
	RadStd		0.38	0.32	0.45	0.39	0.43	0.26	0.40	0.44	0.36	0.43	0.42	0.38

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	+0.19	+0.43	N/A	+0.37	+0.30	+0.30	N/A	N/A	+76	+76

Nomenclature: See separate page